Homework 3

Grant Talbert
February 3, 2026
MA713 Complex Analysis

Boston University

Problem 1. *Define* $\exp(z) = e^z$ *by*

$$e^z = \sum_{n \geq 0} \frac{z^n}{n!}.$$

Prove directly from this definition that $e^{z+w} = e^z e^w$.

Problem 2. *In class we proved that if* $f(z) = \sum_n a_n (z - z_0)^n$ *is a convergent power series on* $B_r(z_0)$, *then* f *is analytic on* $B_r(z_0)$. *Give a one line proof of this in the case* $f = \exp$, $z_0 = 0$, *and* $r = \infty$.

Proof. ??? ■

Problem 3. *Repeat problem 2 but for* $f(z) = 1/(1 - z)$, $z_0 = 0$, *and* $r = 1$.

Proof.

$$\frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n \ \forall z \in B_1(0).$$

■

Problem 4. *Find the complex power series expansion of* $z^2/(1 - z^2)^3$ *about 0 and determine the radius of convergence (do not use Taylor's formula).*

Problem 5. *Give an example of a power series* $\sum_k a_k z^k$ *which converges for every complex value of* z *and which sums to 0 for infinitely many values of* z *but is not the identically zero series.*

Proof. The power series expansion of $\sin(z)$:

$$\sin(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n + 1)!}.$$

This vanishes for $z = n\pi$, $n \in \mathbb{Z}$. This converges everywhere because it is easily shown to be equal to the entire function $(e^{iz} - e^{-iz})/2i$ via Euler's formula. ■

Problem 6. *Give an example of a nonconstant power series* $\sum_k a_k z^k$ *which converges for every* z *but which sums to 0 for no values of* z.

Proof. Power series expansion of $\exp(z)$. Write $z = x + iy$. Then by problem 1,

$$e^z = e^{x+iy} = e^x e^{iy}.$$

Since x, y are purely real, we have that $e^x \neq 0$, and $e^{iy} = \cos y + i \sin y$ by the easy to prove Euler's formula. We then have $\left|e^{iy}\right| = \cos^2 y + \sin^2 y = 1$, hence $e^{iy} \neq 0$. ■